

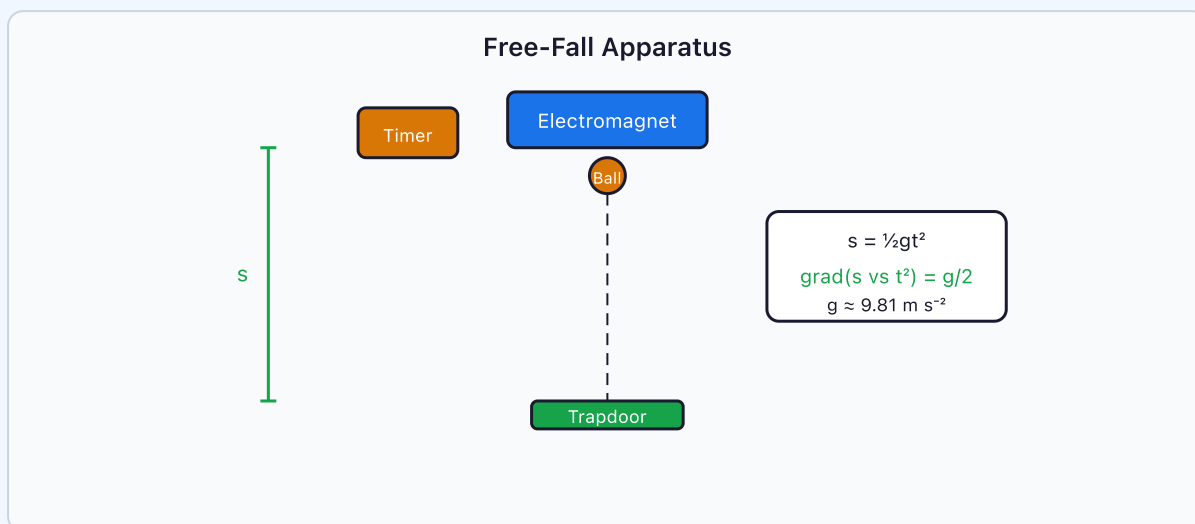
OCR B A-Level Physics — Practical Activities

Practical Endorsement (H557): you must complete at least 12 practical activities that together cover the twelve OCR Practical Activity Groups (PAG1–PAG12). The activities named below are the examples in the specification plus other practicals the specification lists under each topic; your centre may use different activities for the same PAG. The practicals named in the specification can be examined in the written papers.

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- Other practical work (not a PAG): force on a wire; inverse-square law

PAG1 — Investigating Motion: Acceleration of Free Fall



Apparatus

- Electromagnet, steel ball, trapdoor, timer, 2 m ruler, clamp stand

Method

- Set height s . Release ball (timer starts). Ball hits trapdoor (timer stops).
- Repeat 3 times; take mean t . Vary s over 6+ values.
- Plot s vs t^2 . Gradient = $g/2$.

Safety

- Soft pad; stable stand; clear feet

Analysis

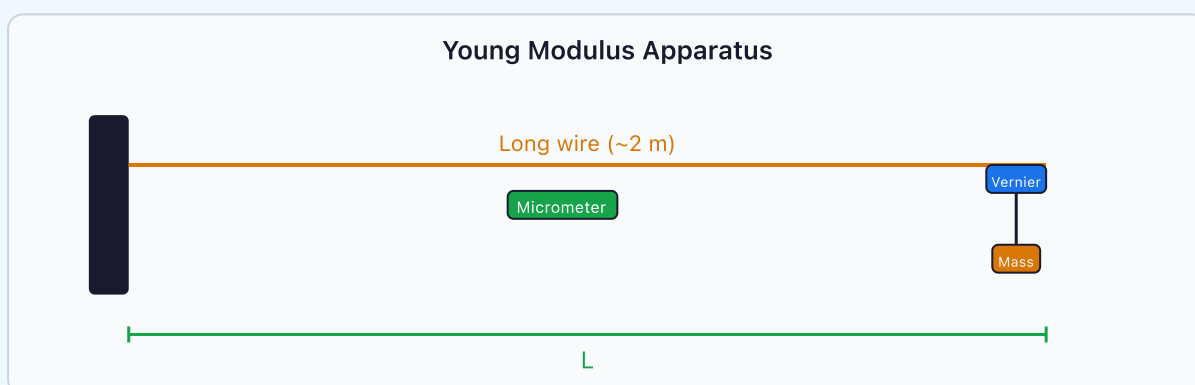
$$s = \frac{1}{2}gt^2 \Rightarrow g = 2 \times \text{gradient}$$

Source of Error	Type	Improvement
Residual magnetism	Systematic	Demagnetiser or mechanical release
Air resistance	Systematic	Small dense ball

Height measurement	Random	Measure bottom of ball to trapdoor
Reaction time (if manual)	Random	Use electronic timer

Exam Tip: A constant timing delay (e.g. the ball taking a moment to release from the magnet) adds the same amount to every t , so s vs t^2 is no longer a straight line (it curves slightly, so the gradient is not $g/2$). Plotting $\sqrt{s}$ against t removes the problem: the line has gradient $\sqrt{g/2}$, so $g = 2 \times \text{gradient}^2$, and its t -intercept reveals the delay. Calculate percentage error from accepted value to evaluate your result.

PAG2 — Investigating Properties of Materials: Young Modulus of a Metal Wire



Apparatus

- Long thin wire (~ 2 m), G-clamp, wooden blocks
- Micrometer, vernier scale or travelling microscope
- Slotted masses, tape measure, safety goggles

Method

1. Measure diameter (6+ readings). Calculate $A = \pi d^2/4$.
2. Measure original length L . Apply small initial load to straighten.

3. Add masses in steps. Record $F = mg$ and extension ΔL .
4. Check elastic behaviour on unloading.
5. Plot stress vs strain. Gradient = Young modulus E .

Safety

- Safety goggles; cushion beneath masses; do not exceed elastic limit

Analysis

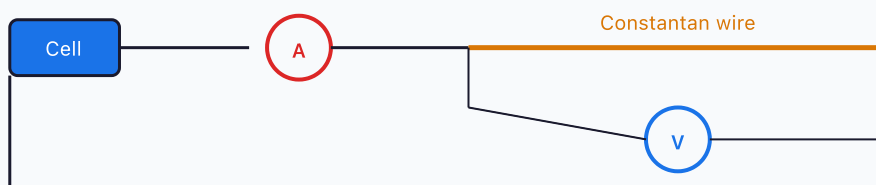
$$E = \sigma/\epsilon = (F/A)/(\Delta L/L)$$

Source of Error	Type	Improvement
Diameter varies	Random	Multiple readings; different positions/orientations
Wire kinks	Systematic	Apply small initial load
Thermal expansion	Systematic	Use reference wire; stable temperature
Small extensions	Random	Long thin wire; travelling microscope

Exam Tip: Percentage uncertainty in d is doubled in the area formula (d^2), making it the dominant source of error. Use a long wire to maximise extensions. The area under the linear stress-strain graph gives strain energy per unit volume.

PAG3 — Investigating Electrical Properties: Resistivity of a Metal

Resistivity Circuit



$$R \text{ vs } L \rightarrow \text{gradient} = \rho/A \rightarrow \rho = \text{gradient} \times A$$

Apparatus

- Constantan wire (~1 m, taped to ruler)
- Ammeter, voltmeter, cell/power supply, switch
- Micrometer screw gauge, metre ruler, crocodile clips

Method

1. Measure wire diameter at 6+ points with micrometer. Calculate $A = \pi d^2/4$.
2. Connect ammeter in series, voltmeter in parallel across wire section.
3. Use clips to select length L . Record V and I . Calculate $R = V/I$.
4. Repeat for 6+ lengths. Plot R vs L .
5. Gradient = ρ/A , so $\rho = \text{gradient} \times A$.

Safety

- Switch off between readings; low voltage; avoid touching hot wire

Analysis

$$R = \rho L/A \Rightarrow \text{gradient of } R \text{ vs } L = \rho/A$$

Expected: $\rho(\text{constantan}) \approx 4.9 \times 10^{-7} \Omega \text{ m}$.

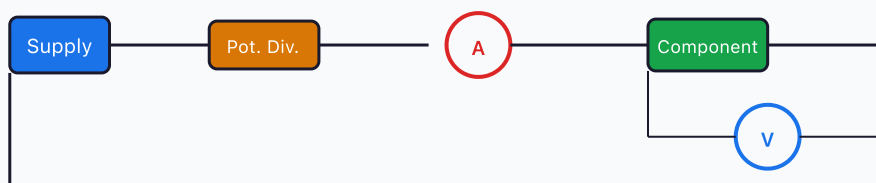
Source of Error	Type	Improvement
Wire heating	Systematic	Switch off between readings; low current

Contact resistance	Systematic	Clean clips; firm connections
Diameter variation	Random	Multiple micrometer readings; use mean
Length measurement at clip	Random	Tape wire to ruler; align carefully

Exam Tip: Constantan is chosen because its resistance barely changes with temperature. The biggest uncertainty typically comes from the diameter (squared in the area calculation). A non-zero y-intercept suggests contact resistance.

PAG3 — Investigating Electrical Properties: I–V Characteristics of Components

V-I Characteristics Circuit



Components: ohmic conductor, filament lamp, diode

Use potential divider for full 0–V range

Apparatus

- DC power supply with potential divider (or variable resistor)
- Ammeter, voltmeter
- Components to test: wire (ohmic), filament lamp, silicon diode
- Switch, connecting wires

Method

1. Connect the potential divider circuit for full voltage range (0 to V_{max}).

2. Place the component under test in series with the ammeter. Voltmeter in parallel across it.
3. Vary voltage from 0 upwards. Record V and I at each setting.
4. For the diode, reverse the connections and repeat.
5. Plot I vs V for each component.

Safety

- Filament lamp gets hot — do not touch
- Low voltage ($< 12\text{ V}$)
- Do not exceed maximum current rating of components

Analysis

- **Ohmic conductor:** Straight line through origin (constant resistance)
- **Filament lamp:** Curve — resistance increases as wire heats up
- **Diode:** Very little current below $\sim 0.6\text{ V}$ (threshold); rapid increase above; negligible current in reverse bias

Source of Error	Type	Improvement
Component heating changes resistance	Systematic	Take readings quickly; allow cooling
Cannot reach 0 V with variable resistor alone	Systematic	Use potential divider circuit
Meter resolution	Random	Use digital meters with appropriate range
Parallax on analogue meters	Random	Read at eye level

Exam Tip: A potential divider allows you to vary the voltage from 0 V upwards, which is essential for accurately determining the diode threshold. A simple variable resistor in series cannot reduce V to 0 V .

PAG3 — Investigating Electrical Properties: EMF and Internal Resistance

EMF and Internal Resistance



$$V = \epsilon - Ir \Rightarrow y\text{-int} = \epsilon, \text{ gradient} = -r$$

Apparatus

- Cell, variable resistor, ammeter, voltmeter, switch

Method

1. Connect cell, ammeter, variable resistor in series. Voltmeter across cell.
2. Record V and I for 6+ settings. Switch off between.
3. Plot V vs I .

Safety

- No short circuits; switch off between readings

Analysis

$$V = \epsilon - Ir. \text{ y-intercept} = \epsilon, \text{ gradient} = -r.$$

Source of Error	Type	Improvement
Cell runs down	Systematic	Work quickly; switch off between readings
Meter resistances	Systematic	Digital meters
Fluctuations	Random	Wait for stable readings

Heating effects

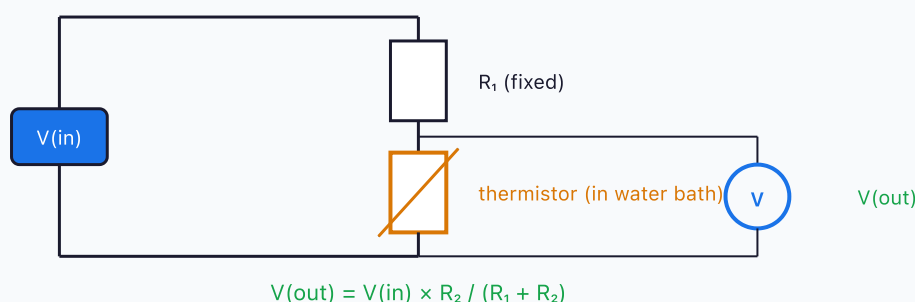
Systematic

Short measurement bursts

Exam Tip: The x-intercept gives the short-circuit current $I = \varepsilon/r$. As the external resistance R increases, V approaches ε (more of the EMF is across the external component).

PAG4 — Investigating Electrical Circuits: Potential Divider Circuits

Potential Divider with a Thermistor



Apparatus

- Low-voltage DC supply or battery, fixed resistors (or a resistance box)
- Sensor: NTC thermistor or LDR; high-resistance digital voltmeter; digital multimeter to measure resistance
- For a thermistor: beaker of water, kettle or heater, thermometer, stirrer (for an LDR: lamp and metre rule)

Method

1. First test the divider with two known resistors: measure V_{in} and V_{out} and check $V_{\text{out}}/V_{\text{in}} = R_2/(R_1 + R_2)$ for several pairs.
2. Replace R_2 with the thermistor, sealed in a thin plastic bag or with insulated leads, and place it in the water with the thermometer.

- Heat the water in steps of about 10 °C. At each step stir, wait for the reading to settle, then record θ and V_{out} .
- Plot V_{out} against θ . Repeat with a different fixed resistor R_1 and compare the sensitivity (change in V_{out} per °C).
- Swap the positions of the sensor and fixed resistor and note that V_{out} now rises instead of falls as temperature increases.

Safety

- Hot water — stable beaker, heat-proof mat, do not touch the heater; keep mains equipment away from spills; low-voltage supply only

Analysis

$$V_{\text{out}} = V_{\text{in}} \times R_2 / (R_1 + R_2) \quad V_1/V_2 = R_1/R_2$$

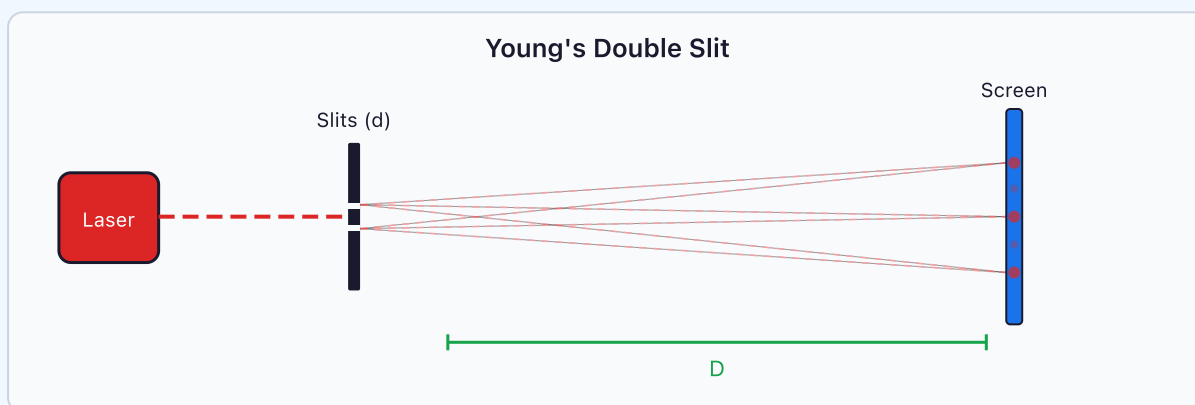
As temperature rises, the NTC thermistor's resistance falls, so the voltage across it falls. The circuit is most sensitive when R_1 is similar to the thermistor's resistance in the middle of the temperature range of interest.

Source of Error	Type	Improvement
Voltmeter draws current (loading)	Systematic	Use a digital voltmeter with resistance much greater than R_2
Thermistor not at water temperature	Systematic	Stir; wait for V_{out} to stabilise before reading
Supply p.d. drops under load	Systematic	Measure V_{in} at every reading, not just once
Thermometer resolution / reading	Random	Digital thermometer; repeat on cooling and average

Exam Tip: Potential difference divides in the same ratio as resistance. If the output is connected to a load (e.g. a meter or another circuit) whose resistance is not much larger

than R_2 , the effective lower resistance falls and V_{out} drops — a common “explain the discrepancy” question.

PAG5 — Investigating Waves: Wavelength by Double Slit and Diffraction Grating



Apparatus

- Laser (or white light + colour filter + single slit)
- Double slit (known separation d)
- Screen, ruler/vernier callipers, tape measure
- Diffraction grating (lines per mm known), protractor or metre rules for the angle
- For sound: signal generator, two loudspeakers, microphone and oscilloscope

Method

1. Darken room. Direct laser through double slit onto distant screen.
2. Measure across several fringe spacings. Divide to get w (fringe spacing).
3. Measure slit-to-screen distance D .
4. Calculate $\lambda = wd/D$.
5. Vary D and check λ remains consistent.

Method (Diffraction Grating)

1. Replace the double slit with a grating, normal to the beam. Calculate the slit spacing $d = 1/(\text{lines per metre})$.
2. Measure the distance from grating to screen and the distance from the central maximum to the 1st, 2nd, ... order maxima on each side.
3. Find each angle θ using $\tan \theta = (\text{distance on screen})/(\text{grating-to-screen distance})$; halve the distance between matching orders on opposite sides to cancel misalignment.
4. Use $n\lambda = d \sin \theta$ for each order and average.

Method (Sound)

1. Drive two loudspeakers a distance d apart from the same signal generator (a few kHz) so they are coherent.
2. Move a microphone along a line a distance D away, watching the oscilloscope trace, and mark successive loud points.
3. Measure the spacing w of the loud points and use $\lambda = wd/D$; check using $v = f\lambda \approx 340 \text{ m s}^{-1}$.

Safety

- Never look into laser beam; "Laser in use" warning
- Remove reflective objects

Analysis

$$\text{Double slit: } \lambda = wd/D \quad \text{Grating: } n\lambda = d \sin \theta$$

Measure across multiple fringes to reduce uncertainty in w . The grating usually gives the more precise result: its maxima are sharp and widely spaced, so the angles can be measured with a small percentage uncertainty.

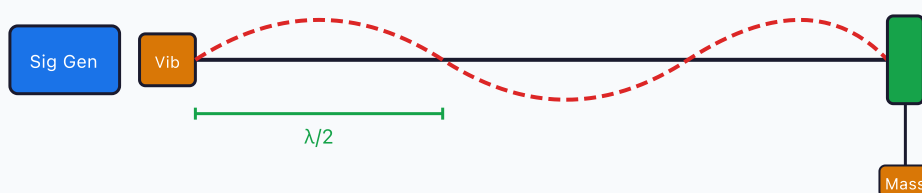
Source of Error	Type	Improvement
Fringes blurred	Random	Use laser for coherent source; darken room
D measurement	Random	Large D reduces % uncertainty

Fringe spacing hard to measure	Random	Measure across 10+ fringes and divide
Laser not perpendicular to slits	Systematic	Align carefully

Exam Tip: For white light, central fringe is white with coloured fringes either side (red outermost because it has the longest wavelength and hence largest fringe spacing). With a laser, all fringes are the same colour and equally spaced.

PAG5 — Investigating Waves: Stationary Waves on a String

Stationary Waves



Apparatus

- Signal generator, vibrator, string, pulley, masses, ruler

Method

1. Set up string between vibrator and pulley with hanging mass.
2. Vary frequency to find fundamental and harmonics. Record f and measure L .
3. Vary tension. Plot f vs $1/L$ or f vs $\sqrt{T}$.

Safety

- Safety screen; soft pad beneath masses

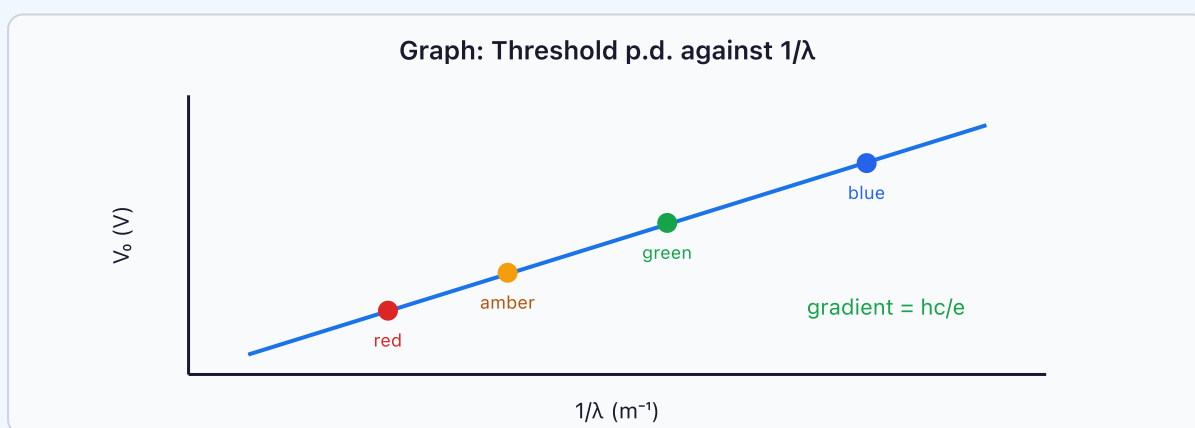
Analysis

$$f = (1/2L)\sqrt{(T/\mu)}$$

Source of Error	Type	Improvement
Resonance hard to judge	Random	Paper rider method
Length measurement	Random	Measure node to node
Hanger mass omitted	Systematic	Include in tension
String not uniform μ	Random	Weigh measured length

Exam Tip: The fundamental frequency has one antinode. Each harmonic adds one more antinode. The frequency of the n th harmonic is n times the fundamental frequency.

🔧 PAG6 — Investigating Quantum Effects: Planck Constant Using LEDs



Apparatus

- LEDs of several colours with known peak wavelengths (data sheet, or measured with a diffraction grating)

- Protective resistor (a few hundred Ω) in series with each LED
- Low-voltage supply with potential divider, digital voltmeter, milliammeter
- Short black tube or card shield to view the LED in dim light

Method

1. Connect the LED (correct polarity) in series with the protective resistor and milliammeter, fed from the potential divider. Put the voltmeter directly across the LED.
2. Darken the room. Starting from 0 V, raise the p.d. slowly until the LED just begins to glow when viewed down the tube. Record this threshold p.d. V_0 .
3. Alternatively record the I–V characteristic and extrapolate the straight, conducting part of the curve back to $I = 0$ to find V_0 .
4. Repeat each reading (approaching from below each time) and take a mean. Repeat for every LED.
5. Plot V_0 against $1/\lambda$.

Safety

- Always include the protective resistor so the LED current stays within its rating; do not stare into bright or blue/UV LEDs at close range

Analysis

$$eV_0 \approx hf = hc/\lambda \Rightarrow V_0 = (hc/e) \times (1/\lambda)$$

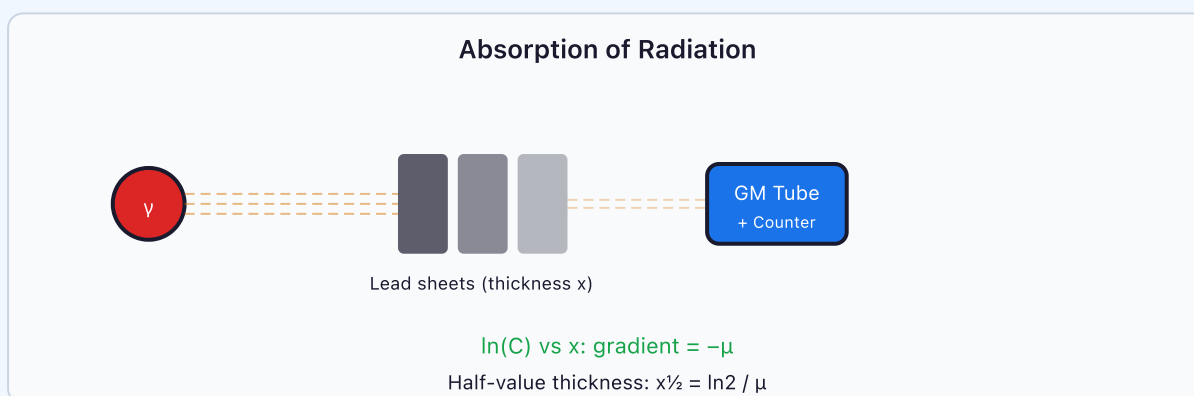
$h = \text{gradient} \times e / c$. Expected: $h \approx 6.6 \times 10^{-34} \text{ J s}$. At threshold, each electron crossing the junction gives up energy eV_0 , which is emitted as one photon.

Source of Error	Type	Improvement
Judging when the LED "just" lights	Random	Dark room, viewing tube, repeat and average; or use the I–V extrapolation
LEDs emit a spread of wavelengths	Systematic	Use the peak wavelength from the data sheet or measure with a grating

Energy lost in the junction and leads	Systematic	Use the gradient, not individual points, so a constant offset does not affect h
Voltmeter across LED and resistor	Systematic	Connect the voltmeter across the LED alone

Exam Tip: Use the gradient of V_0 against $1/\lambda$ rather than a single LED: any constant offset in the threshold p.d. only shifts the intercept. Remember to convert wavelengths from nm to m before calculating $1/\lambda$.

PAG7 — Investigating Ionising Radiation: Absorption of Radiation



Apparatus

- Gamma source, GM tube + counter, lead sheets of known thickness, callipers

Method

1. Measure background count rate (no source, 60+ s).
2. Fix source-detector distance. Record count rate with no absorber.
3. Insert lead sheets of increasing total thickness. Record corrected count rate.
4. Repeat for 6+ thicknesses.

5. Plot $\ln(\text{corrected count rate})$ vs thickness x .

Safety

- Tongs; distance; time minimised; ALARA; lead storage

Analysis

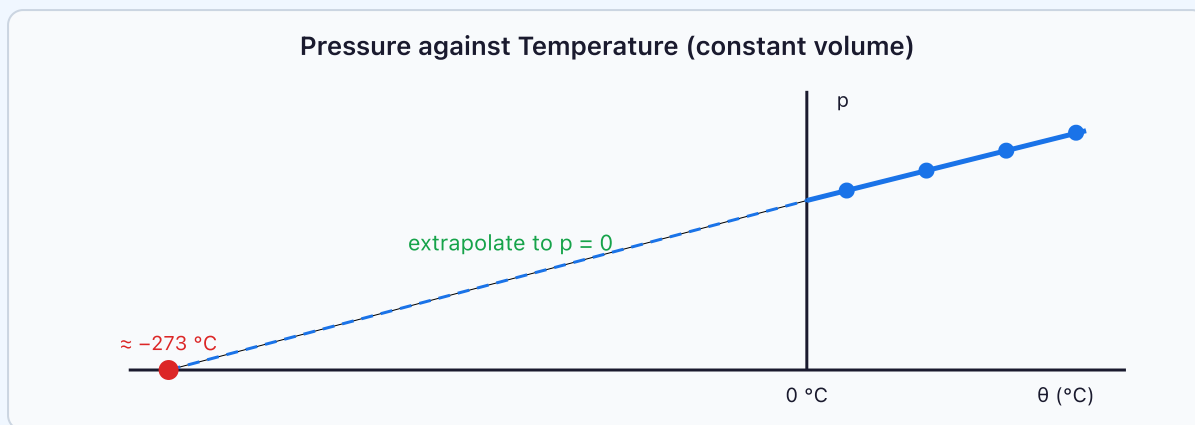
$$I = I_0 e^{-\mu x} \Rightarrow \ln(C) \text{ vs } x \text{ gives gradient} = -\mu$$

Half-value thickness: $x_{1/2} = \ln 2 / \mu \approx 0.693 / \mu$.

Source of Error	Type	Improvement
Random nature of decay	Random	Long count times; multiple readings
Background fluctuations	Random	Long background count; multiple measurements
Absorber thickness not uniform	Systematic	Measure at multiple points with callipers
Scattered radiation	Systematic	Collimate the beam

Exam Tip: Different types of radiation have very different absorption characteristics. Alpha is stopped by paper, beta by a few mm of aluminium, and gamma requires several cm of lead. The half-value thickness is specific to both the radiation energy and the absorber material.

PAG8 — Investigating Gases: Estimating Absolute Zero from Pressure and Temperature



Apparatus

- Round-bottomed flask of dry air, sealed with a bung and short tube to a pressure gauge (Bourdon gauge) or pressure sensor and data logger
- Large beaker or water bath, heater, thermometer, stirrer, ice

Method

1. Clamp the flask so it is fully submerged in the bath, with only the short connecting tube above the water.
2. Start with ice-cold water. Stir, wait until the pressure reading is steady, then record θ and p .
3. Heat in steps of about 10 °C up to around 80 °C, stirring and waiting for a steady reading each time.
4. Plot p (absolute pressure, not gauge pressure) against θ in °C and draw the line of best fit.
5. Extrapolate the line back to $p = 0$ and read off the temperature intercept.

Safety

- Hot water — heat-proof mat, stable clamp; check the flask for cracks and do not heat a sealed flask far above 100 °C; eye protection

Analysis

$$p \propto T \text{ at constant volume} \Rightarrow p = 0 \text{ when } T = 0 \text{ K}$$

θ -intercept = $-(p \text{ at } 0 \text{ °C})/\text{gradient}$. Expected: about -273 °C . This is the basis of the absolute (kelvin) scale: $T/\text{K} = \theta/\text{°C} + 273$.

Source of Error	Type	Improvement
Air in the tube is not at bath temperature	Systematic	Keep the connecting tube short and narrow
Gas not in thermal equilibrium with water	Systematic	Stir; wait for a steady pressure before reading
Long extrapolation magnifies gradient uncertainty	Random	Wide temperature range; many points; worst-fit lines to estimate uncertainty
Gauge reads relative to atmosphere	Systematic	Add atmospheric pressure to gauge readings

Exam Tip: The intercept is found by extrapolating a long way beyond the data, so a small change in gradient moves it a lot. Real gases liquefy before 0 K, so absolute zero is an extrapolation of ideal-gas behaviour, not a temperature the gas actually reaches.

PAG8 — Investigating Gases: Pressure and Volume (Boyle's Law)

Boyle's Law Apparatus



P vs $1/V$: straight line through origin

$PV = \text{constant}$ at constant T

Apparatus

- Sealed syringe or Boyle's law apparatus, pressure gauge, thermometer

Method

1. Record initial V and P.
2. Change V slowly. Wait for temperature equilibrium.
3. Record V and P. Repeat for 6+ data points.
4. Plot P vs 1/V.

Safety

- Do not over-pressurise; safety screen

Analysis

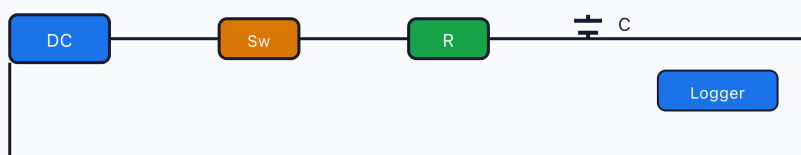
$PV = \text{constant} \Rightarrow P \text{ vs } 1/V \text{ is linear through origin}$

Source of Error	Type	Improvement
Temperature rise on compression	Systematic	Compress slowly; wait for equilibrium
Piston friction	Systematic	Lubricate; tap gently
Gas leaks	Systematic	Check seals
Gauge parallax	Random	Read at eye level; digital gauge

Exam Tip: Boyle's law only holds at constant temperature. If the graph of P vs 1/V does not pass through the origin, this suggests a systematic error (e.g. trapped air of unknown volume, or temperature change).

PAG9 — Investigating Capacitors: Charge, Discharge and Time Constant

RC Discharge Circuit



$$V = V_0 e^{-t/RC} \Rightarrow \ln(V) \text{ vs } t: \text{gradient} = -1/RC$$

Apparatus

- Capacitor (470–2200 μF), resistor, DC supply, two-way switch, datalogger

Method

- Charge capacitor through resistor. Record V vs t .
- Discharge. Record V vs t .
- Plot $\ln(V)$ vs t for discharge. Gradient = $-1/RC$.

Safety

- Correct polarity; discharge before handling; low voltage

Analysis

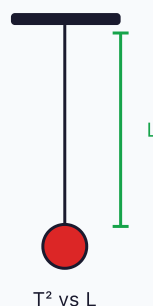
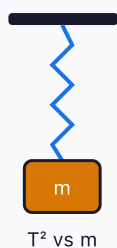
$$V = V_0 e^{-t/RC} \Rightarrow \ln V = \ln V_0 - t/RC$$

Source of Error	Type	Improvement
Voltmeter draws current	Systematic	High-impedance datalogger
Manual timing	Random	Datalogger
Capacitor leakage	Systematic	High-quality capacitor
Component tolerance	Systematic	Measure actual R and C independently

Exam Tip: The time constant $\tau = RC$. After 1τ , V falls to 37% of V_0 . After 5τ , $V < 1\%$ of V_0 (effectively fully discharged). Choose R and C so τ is 10–30 s for manageable timing.

PAG10 — Investigating Simple Harmonic Motion: Mass-Spring and Pendulum

SHM Investigations



Both graphs: straight line through origin

Apparatus

- Spring, masses, clamp stand, stopwatch, ruler
- String, bob, protractor, ruler

Method (Mass-Spring)

1. Hang mass m . Displace slightly, release. Time 10 oscillations.
2. Repeat for 6+ masses. Plot T^2 vs m .

Method (Pendulum)

1. Measure L . Displace $<10^\circ$, release. Time 10 oscillations.
2. Repeat for 6+ lengths. Plot T^2 vs L .

Safety

- Stable stand; clear of swinging masses

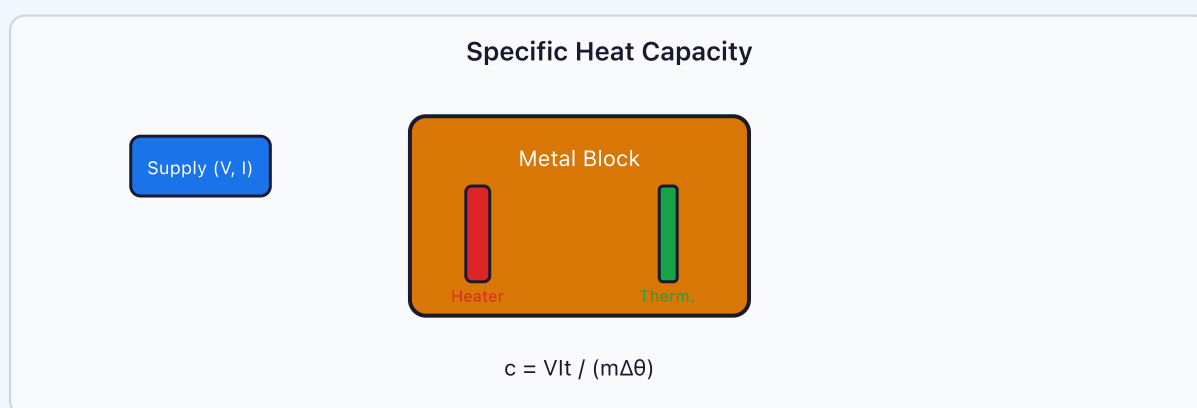
Analysis

Spring: $T^2 = (4\pi^2/k)m$ Pendulum: $T^2 = (4\pi^2/g)L$

Source of Error	Type	Improvement
Miscounting oscillations	Random	Fiducial marker; count from equilibrium
Large amplitude (pendulum)	Systematic	Keep $\theta < 10^\circ$
Reaction time	Random	Time 10+ oscillations
Spring mass	Systematic	Add $m_{\text{spring}}/3$ as effective mass

Exam Tip: The pendulum period is independent of mass. The spring period is independent of g. Plotting T^2 (not T) linearises the graph. The gradient gives k or g respectively.

PAG11 — Investigation: Specific Heat Capacity of a Material



Apparatus

- Metal block with holes, heater, thermometer
- Ammeter, voltmeter, power supply, stopwatch, balance, insulation

Method

1. Weigh block. Insert heater and thermometer with oil.
2. Record initial θ . Heat for measured time t . Record V , I , final θ .
3. $c = VIt / (m\Delta\theta)$.

Safety

- Hot block; tongs/gloves; heat mat; heater fully inserted

Analysis

$$c = VIt / (m\Delta\theta)$$

Source of Error	Type	Improvement
Heat loss	Systematic (c too high)	Insulate block
Poor thermal contact	Systematic	Oil in thermometer hole
V/I fluctuations	Random	Regulated supply; monitor throughout
Thermometer lag	Systematic	Wait for equilibrium; stir if liquid

Exam Tip: The measured c is always slightly too high because of heat losses. Insulation and a cooling correction can improve accuracy. Plotting $\Delta\theta$ vs t gives a straight line whose gradient = $VI/(mc)$.

PAG12 — Research Skills

PAG12 is not a bench experiment. You research a physics topic using online and offline sources, then produce a short report or presentation that explains the physics and correctly cites where the information came from. Example topics include how the Global Positioning System (GPS) works and how radioactive tracers are used in medical imaging.

What You Do

1. Turn the topic into focused questions (e.g. "How does a receiver work out its distance from a satellite?").
2. Search more than one type of source: textbooks, reputable websites (universities, scientific bodies, government agencies), and journal or magazine articles.
3. Make notes in your own words and record full reference details as you go.
4. Check key facts against a second independent source; note any disagreement.
5. Write up the physics clearly, with calculations or diagrams where helpful, and a reference list.

Citing Sources

- Use one consistent style (e.g. author, year, title, publisher or URL, date accessed).
- Put a numbered or author–year citation in the text wherever you use information from a source.
- Quote only short phrases, in quotation marks, and paraphrase everything else.

Example Topics: The Physics to Include

- **GPS:** each satellite broadcasts its position and a precise time from an atomic clock. The receiver measures each signal's travel time and calculates distance = $c \times t$. Signals from at least four satellites are needed: three distances fix the position and the fourth corrects the receiver's less accurate clock. An error of 1 μs in timing gives about 300 m error in distance.
- **Medical tracers:** a gamma emitter such as technetium-99m (half-life about 6 hours) is attached to a chemical taken up by the organ of interest and detected outside the body with a gamma camera. Gamma rays are used because they escape the body and are weakly ionising; the short half-life limits the patient's dose while still allowing time for the scan.

Judging Sources

Source	Strengths	Watch out for
Textbook	Reviewed; pitched at the right level	May be out of date for new technology

University / scientific body website	Expert, usually current	May be too advanced; check the date
Manufacturer or company site	Real specifications and data	Possible commercial bias
Open-edit encyclopaedia or forum	Good starting point for search terms	Verify facts elsewhere; cite the original source instead

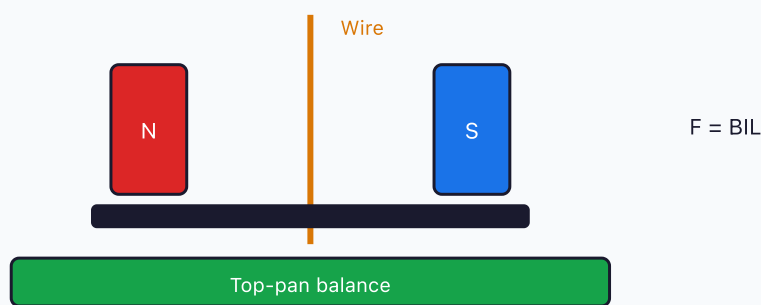
Exam Tip: Research skills can also be tested in written papers (especially the scientific literacy paper, which is based on an advance notice article): be ready to comment on the reliability of a source, to spot where a claim needs evidence, and to explain why data from two sources might differ.

Other Practical Work (not a Practical Activity Group)

These are not PAG activities for H557. The force on a current-carrying wire is listed as a practical activity in the magnetism content, so it can still be examined; the inverse-square law for gamma rays is not in the H557 content but gives useful extra practice with ionising-radiation technique.

Force on a Current-Carrying Wire (flux density with a digital balance)

Force on Current-Carrying Wire



Apparatus

- U-magnet, wire cradle, top-pan balance, DC supply, ammeter, ruler

Method

1. Magnet on balance. Zero. Position wire in field.
2. Vary I . Record I and Δm . $F = \Delta m \times g$.
3. Measure wire length L in field. Plot F vs I .
4. Gradient = BL , so $B = \text{gradient}/L$.

Safety

- Low current; short bursts; magnets away from electronics

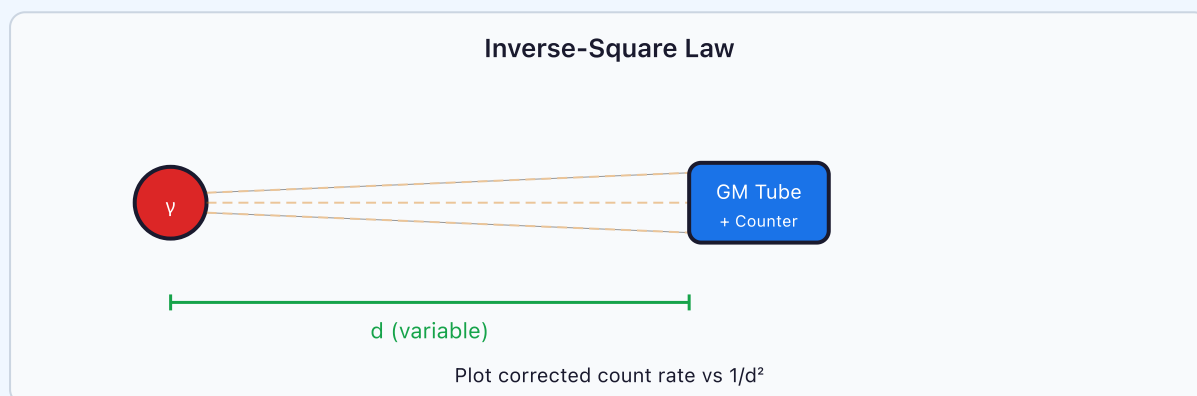
Analysis

$$F = BIL \Rightarrow \text{gradient of } F \text{ vs } I = BL$$

Source of Error	Type	Improvement
Draughts/vibrations	Random	Draught shield; stable surface
Non-uniform field	Systematic	Strong flat-faced magnets; wire at centre
Wire not perpendicular	Systematic	Careful alignment
Wire heating	Systematic	Short bursts of current

Exam Tip: The balance reading changes because by Newton's third law, the force on the wire is equal and opposite to the force on the magnet. $F = BIL$ only when B , I , and L are mutually perpendicular.

Inverse-Square Law for Gamma Radiation (extra practice)



Apparatus

- Gamma source, GM tube + counter, metre ruler, source holder

Method

1. Measure background (60+ s, no source).
2. Place source at distance d . Count for 60 s. Subtract background.
3. Repeat for 6+ distances. Take 3 readings at each.
4. Plot corrected count rate C vs $1/d^2$.

Safety

- Tongs; maximise distance; minimise time; ALARA; lead-lined storage

Analysis

$$I \propto 1/d^2 \Rightarrow C \text{ vs } 1/d^2 \text{ should be linear through origin}$$

Or plot $\ln(C)$ vs $\ln(d)$: gradient should be -2 .

Source of Error	Type	Improvement
Random decay	Random	Count for longer; more repeats
Background fluctuations	Random	Long background count
Distance to tube centre	Systematic	Measure to GM tube window
Non-point source	Systematic	Use large d values

Exam Tip: Always subtract background before plotting. If the graph curves at small d , the point-source approximation breaks down. The inverse-square law applies because gamma radiation spreads out equally in all directions from a point source.